Intention Understanding

Intention Understanding
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1. Course Content

Basic: Master how to customize a user's unique intention understanding function through a RAG knowledge base (which can be understood as adding training samples to the knowledge base to provide context reference for the AI model's retrieval).

Advanced: Master how to debug the intention understanding effect on the Dify side.

2. Preparation

2.1 Configuring the Intention Mapping File

- The file here is used to store personal vague intentions and the corresponding task operations that the robot is expected to perform.
- Open the example file in this lesson's folder. Following the reference format, you can add multiple personally customized intentions. The following is a simple example:

query	answer
I am a bit thirsty	1. Navigate to the kitchen, 2. Check whether there is mineral water or a drink, 3. If so, grasp the drink with the robotic arm, 4. Navigate back to the starting position

2.2 Configuring the Knowledge Base

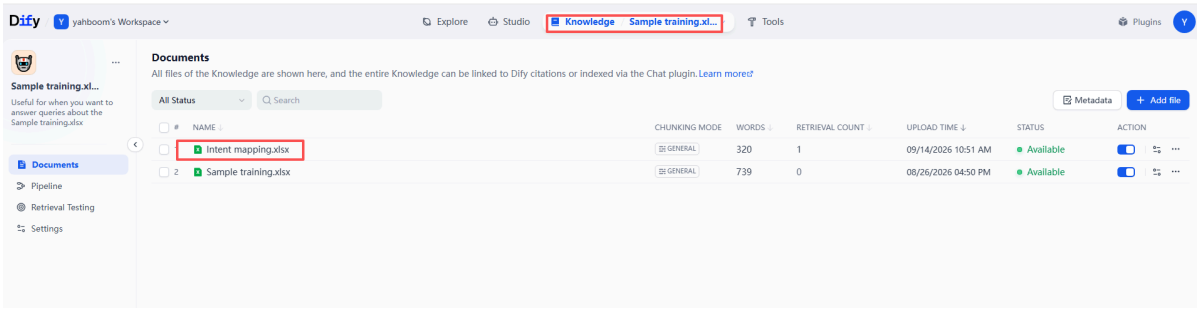
- Then you need to upload the edited intention mapping file to the `RAG` knowledge base of `Dify`. The example file **Intent mapping.xlsx** is in the **Example Files** folder of this lesson.

query	answer
I'm a little thirsty	1. Navigate to the kitchen. 2. Check if there is any mineral water or drink
我感觉有点口渴了	1. 移动至厨房。 2. 检查是否有矿泉水或饮料。 3. 若有，使用机械臂拾取饮料。 4. 导航

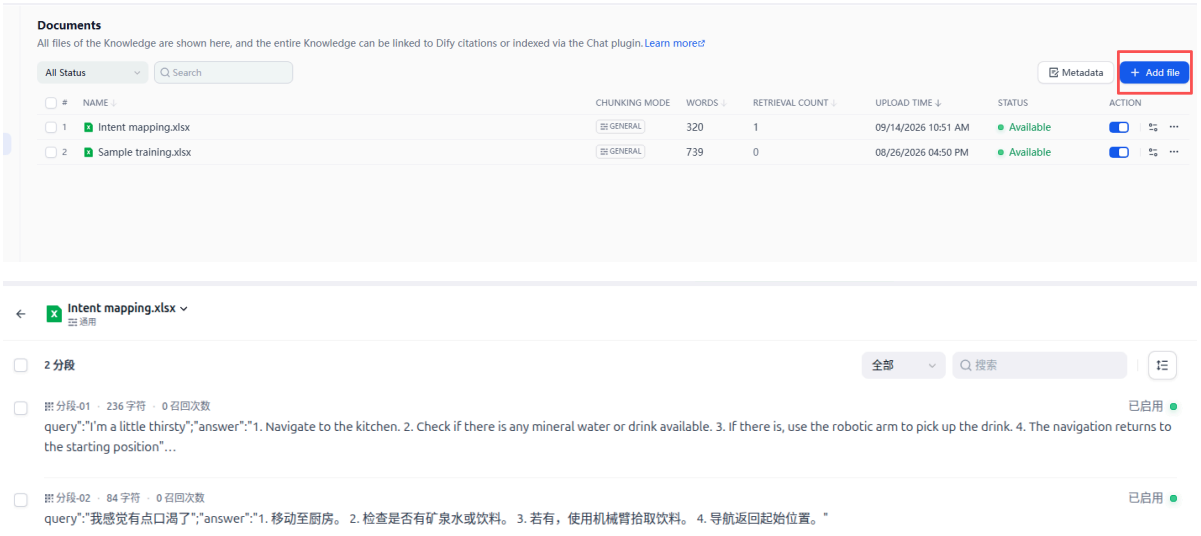
> **Tip**

> - For detailed operations on the RAG knowledge base, refer to the tutorial in "2. AI Model Development — 04-Deploying the RAG Knowledge Base".

- Dify presets a reference knowledge base named `Intent mapping` at the factory to demonstrate how to use the intention understanding function.



- You can modify the `Intent mapping.xlsx` template, or you can directly delete it and add your own file.



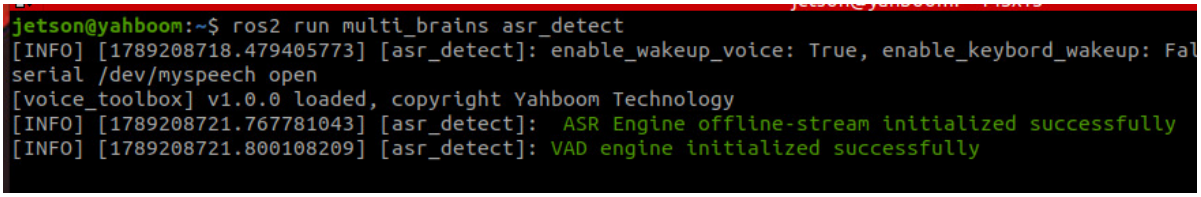
> **Tip**
>
> Note:
>
> - To ensure the best intention understanding effect, it is recommended to set the `Intent mapping` intention understanding knowledge base to high-quality mode, because intention understanding often needs to retrieve relevant segments between similar semantic hints.

- For the operation steps of recording semantic locations to the map mapping file, refer to the previous lesson "Multimodal Visual Understanding + SLAM Navigation".

```
ros2 launch yahboomcar_bringup dify_llm_agents.launch.py use_nav:=true
use_camera:=true use_tts:=true
```

- Voice recognition node

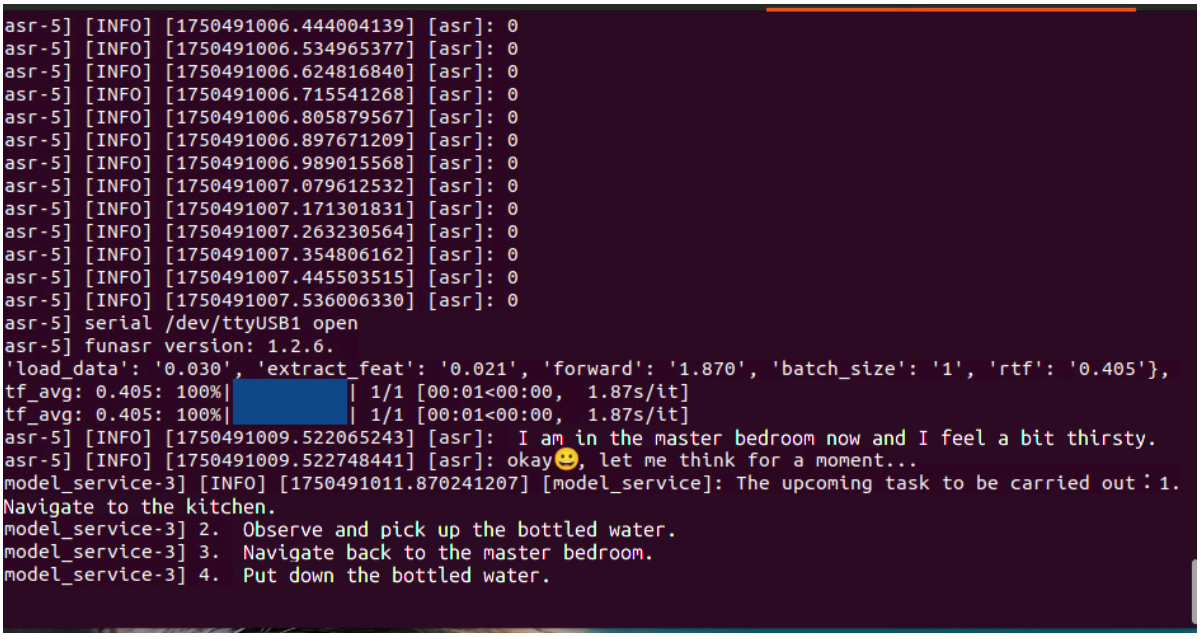
```
ros2 run multi_brains asr_detect
```



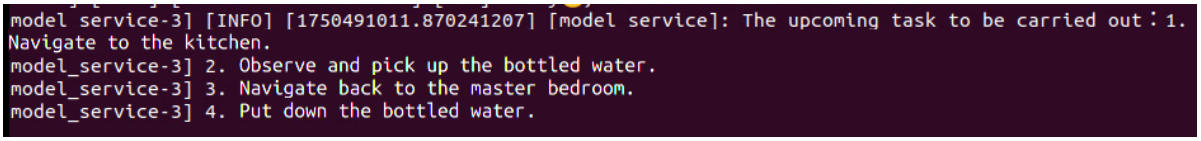
3. Running the Cases

The test cases here are for reference only. You can write your own conversation commands as you like.

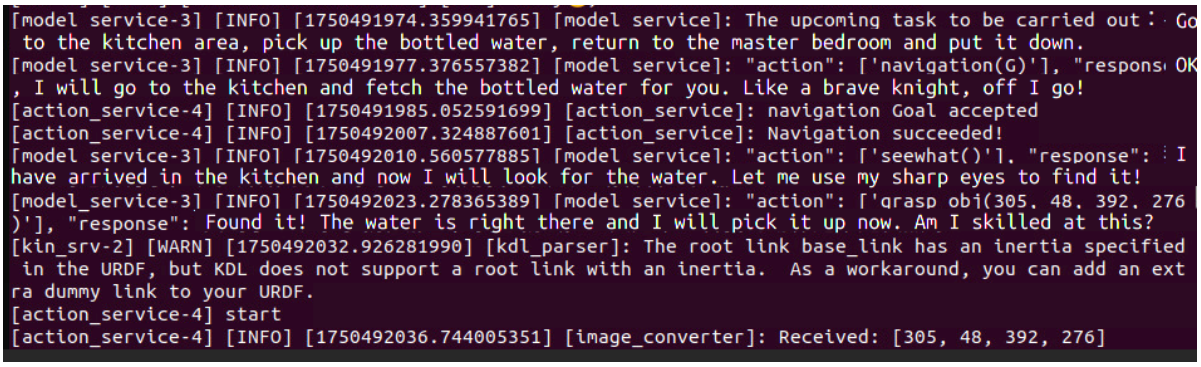
- I am in the master bedroom now, and I feel a bit thirsty.



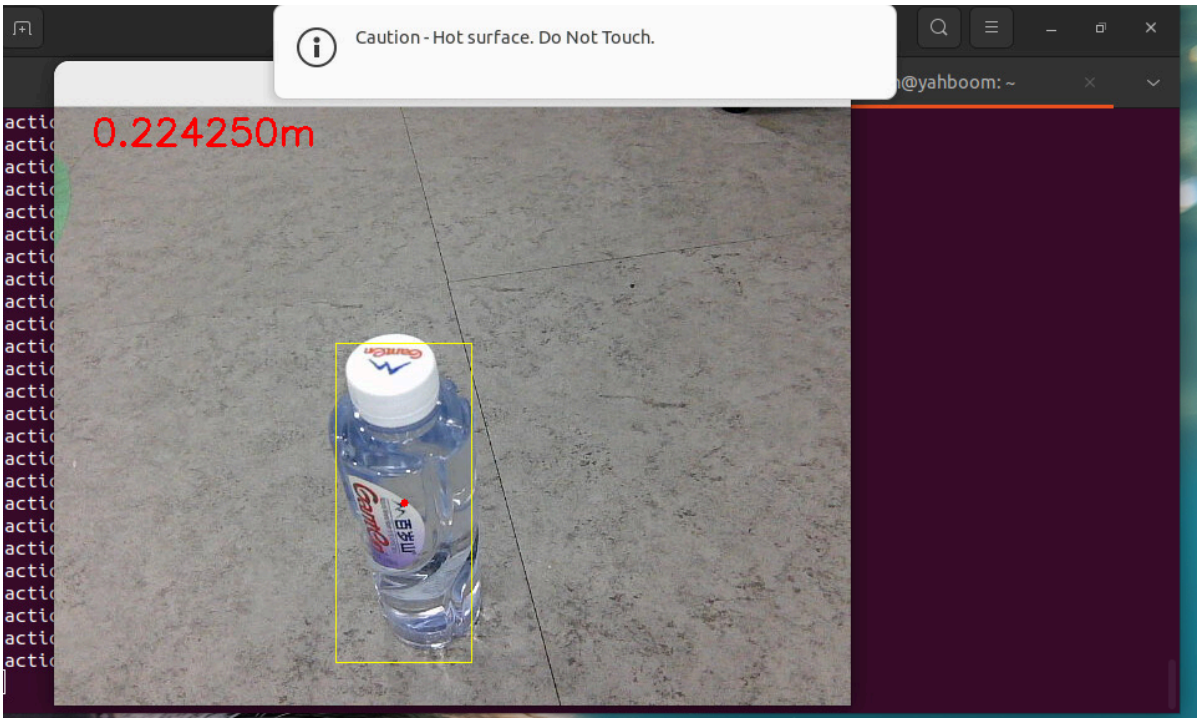
The decision-layer model plans the task steps:



According to the task steps planned by the decision-layer large model, execute them in sequence:



When the robotic arm grasps, a visualization window is displayed, as shown below:



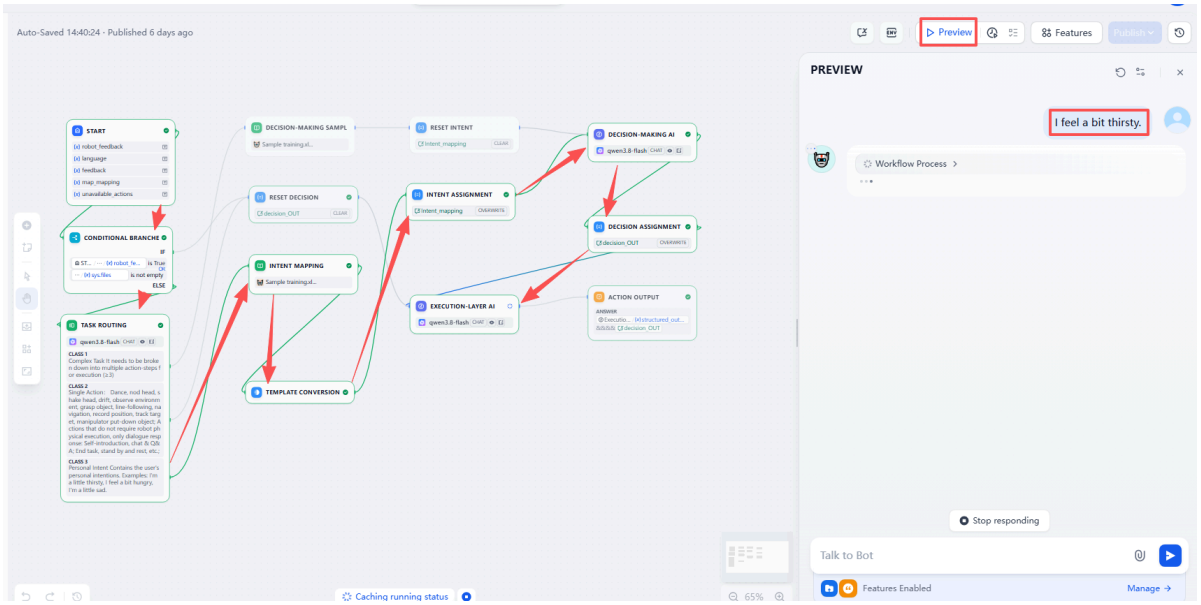
After reaching the "master bedroom", the robot uses the robotic arm to put down the red block in its hand and prompts the user that the task is complete.

```
 in the URDF, but KDL does not support a root link with an inertia.  As a workaround, you can add an ext
ra dummy link to your URDF.
[kin_srv-2] [WARN] [1750492056.766044511] [kdl_parser]: The root link base_link has an inertia specified
 in the URDF, but KDL does not support a root link with an inertia.  As a workaround, you can add an ext
ra dummy link to your URDF.
[kin_srv-2] [WARN] [1750492056.774184364] [kdl_parser]: The root link base_link has an inertia specified
 in the URDF, but KDL does not support a root link with an inertia.  As a workaround, you can add an ext
ra dummy link to your URDF.
[kin_srv-2] [WARN] [1750492056.781544546] [kdl_parser]: The root link base_link has an inertia specified
 in the URDF, but KDL does not support a root link with an inertia.  As a workaround, you can add an ext
ra dummy link to your URDF.
[kin_srv-2] [WARN] [1750492056.789258466] [kdl_parser]: The root link base_link has an inertia specified
 in the URDF, but KDL does not support a root link with an inertia.  As a workaround, you can add an ext
ra dummy link to your URDF.
[kin_srv-2] [WARN] [1750492056.796697211] [kdl_parser]: The root link base_link has an inertia specified
 in the URDF, but KDL does not support a root link with an inertia.  As a workaround, you can add an ext
ra dummy link to your URDF.
[model service-3] [INFO] [1750492065.414920973] [model service]: "action": ['navigation(A)'], "response"
: The water is safely in my hand and I am heading back to the master bedroom. A loyal knight, I deliver!
[action_service-4] [INFO] [1750492074.955772124] [action_service]: navigation Goal accepted
[action_service-4] [INFO] [1750492092.422674757] [action_service]: Navigation succeeded!
[model service-3] [INFO] [1750492095.280458636] [model service]: "action": ['putdown()'], "response": 我
I am back in the master bedroom and will put the water here. Enjoy it, I hope it quenches your thirst!
[model service-3] [INFO] [1750492111.980638518] [model service]: "action": ['finishtask()'], "response":
 Task complete! If you need any more help, just tell me. Have a wonderful day!
```

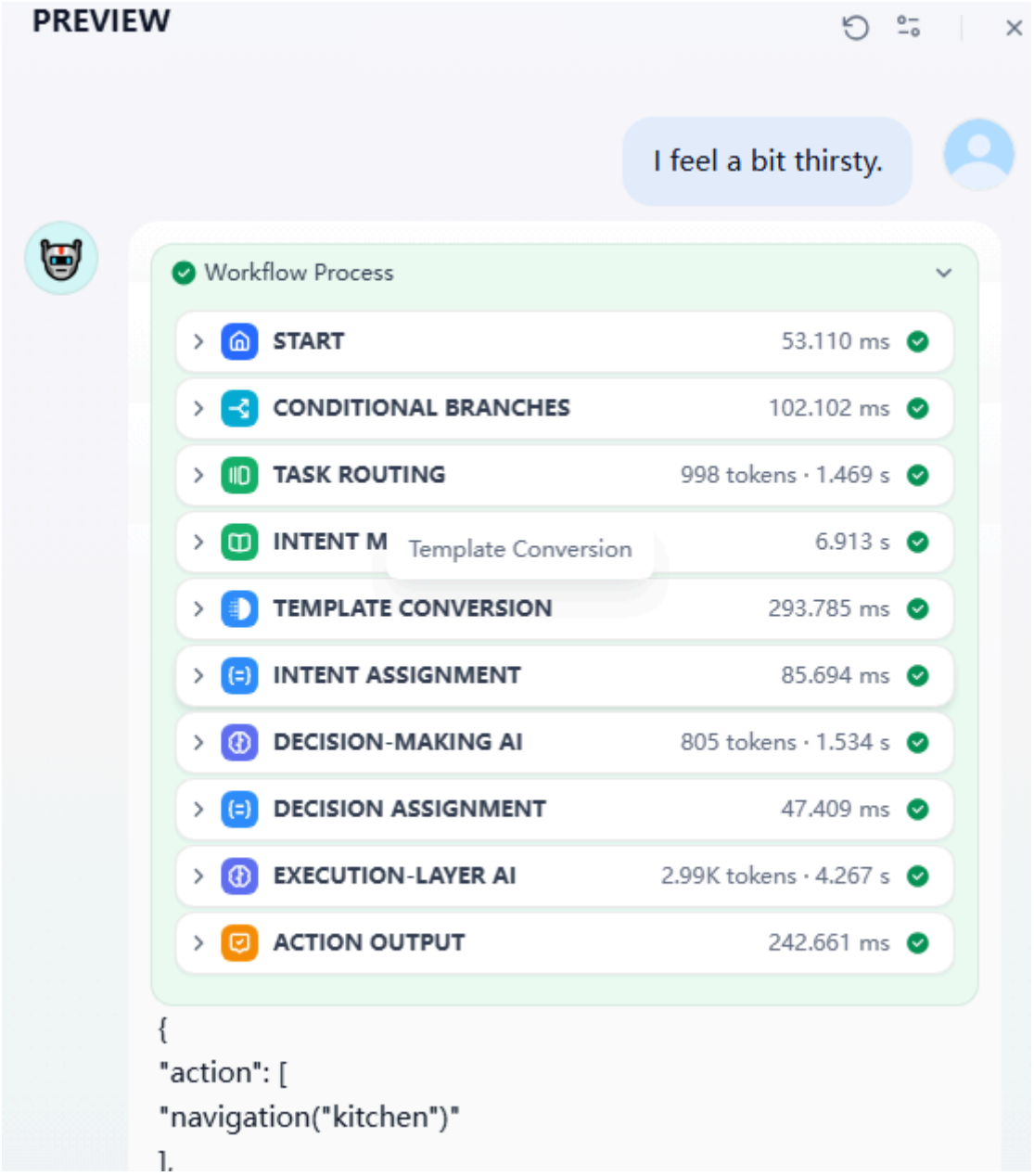
4. Effect Debugging

4.1 Visual Workflow

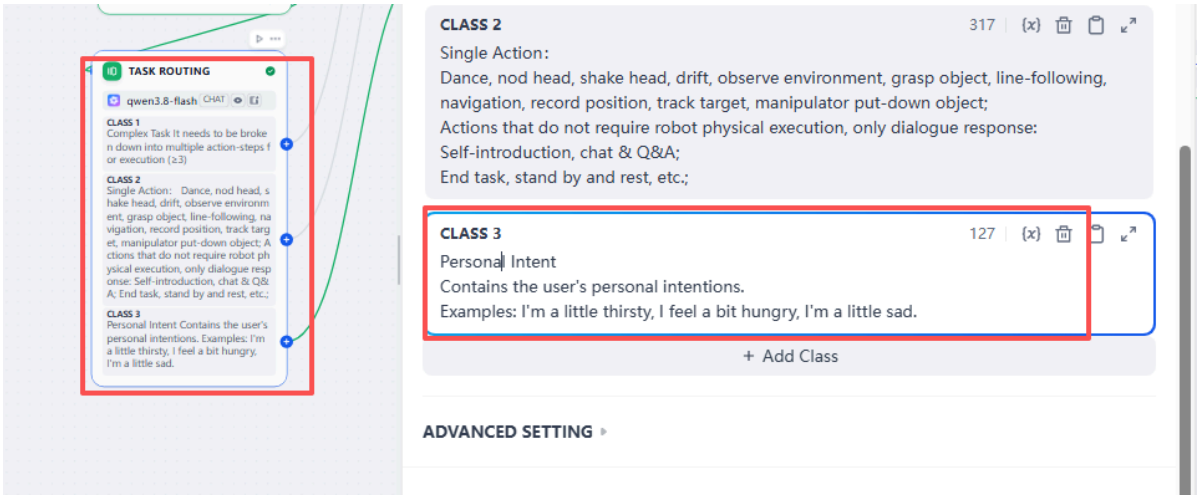
- Open the multi_brains application, then click preview. You can enter test content to view the data flow.



- You can also open the workflow to view the time consumption of each step and the input and output content of each process.



- If in some contexts the classification given by the task routing module is inaccurate and is not classified into `Category 3: Personal Intent`, you can add restrictions on the personal intent here or supplement the context description.



- In addition, you can also separately test the retrieval effect of the input sentence in the knowledge base of the intent mapping.

